

Technical Review: Entropy-Gated Contrastive Decoding

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Overall assessment

The manuscript presents a potentially useful empirical idea: route uncertain binary relation queries to a text-only contrastive decoder while leaving other queries on the ordinary greedy path. The equations defining the standard distribution, Shannon entropy, the adaptive plausibility set, and the contrastive logits are mutually consistent; in particular, the stated proportionality

$$P_{\text{std}}(y)^{1+\alpha}/P_{\text{cf}}(y)^{\alpha}$$

follows from the logit formula up to a token-independent normalization constant. I found no dimensional, sign, inverse-trigonometric, quadrant, or coordinate-transform error in those equations.

The paper nevertheless requires major revision before its main scientific claims are supported. The experiments show that a supervised, per-model entropy threshold can select a useful subset of one-token decisions. They do not show that entropy detects “Visual Inertia,” that the selected models are “structurally sound,” or that the intervention restores visual grounding. A definite held-out class-count inconsistency, missing implementation details at the exact one-token decision point, and statistical tests misaligned with the headline metric also prevent confident interpretation of the reported gains.

Major technical findings

1. The experiment does not identify Visual Inertia or vision–text conflict

Location. Abstract; Introduction; manuscript Sections 3–5, especially the statements that an entropy spike “signals conflict between visual evidence and the language prior,” that cross-attention stagnates on the evaluated models, and that EGCD “recovers suppressed visual evidence”; Conclusion.

Problem. Unsupported claim. The measured signal is full-vocabulary first-token entropy. Entropy can reflect lexical uncertainty, calibration, prompt formatting, or confident/uncertain label bias; it is not a measure of disagreement between the multimodal and text-only passes. More importantly, the manuscript states that the binary evaluation uses $T = 1$. The claimed Visual Inertia mechanism is a token-wise attention trajectory in which attention settles and grounding decays as decoding proceeds, but the experiment contains no generated-token trajectory beyond the answer token and reports no attention measurement. Error-prediction AUCs of 0.62–0.79 show association with correctness, not association with Visual Inertia or evidence that high-entropy items specifically benefit from contrastive decoding. The manuscript itself leaves the required stratified benefit analysis and attention analysis to future work.

Why it matters. The empirical result may support uncertainty-gated contrastive classification, but the title, abstract, contribution list, Results headings, and conclusion repeatedly give it a causal

grounding interpretation that the design cannot establish. The same performance pattern could arise from confidence-dependent Yes/No bias without any attention stagnation.

Suggested fix. Either (i) reframe the contribution throughout as a model-calibrated, first-token uncertainty gate for binary relation classification, treating Visual Inertia only as a motivation, or (ii) add direct evidence: token-wise cross-attention or visual-sensitivity trajectories, multimodal-counterfactual divergence, and a label-stratified analysis of when CD fixes versus harms predictions as a function of entropy. A multi-token or open-ended evaluation is needed to support claims about grounding decay “as decoding proceeds.” Remove or qualify “structurally sound,” “real trouble,” “recovers visual evidence,” and “vision-text conflict” unless those properties are measured.

2. The stated held-out class counts contradict every reported percentage

Location. Experimental Setup, Section 4.4, versus manuscript Table 1.

Problem. Definite error. The text says that the 842 held-out queries comprise 488 positive and 354 negative examples. The table percentages instead align, to rounding precision, with 483 positive and 359 negative examples. For example,

$$\frac{227}{483} = 46.9979\% \approx 47.00\%, \quad \frac{339}{359} = 94.4290\% \approx 94.43\%,$$

which reproduce the Qwen2-VL-2B greedy Recall and Specificity. Likewise, $223/483 = 46.17\%$ and $330/359 = 91.92\%$ reproduce the Phi-3.5-Vision row. The stated denominators 488 and 354 cannot reproduce these values at the shown precision.

Why it matters. The class counts determine Recall, Specificity, balanced accuracy, uncertainty intervals, and any label-stratified analysis. The contradiction also raises uncertainty about which split was actually used for threshold selection and testing.

Suggested fix. Correct the text or regenerate the table from the declared split. Release the exact query IDs and labels, state that this is a group split by image ID (rather than the unclear phrase “stratified by image id”), and generate all counts and metrics from one version-controlled results file. Report integer confusion-matrix counts alongside percentages.

3. The one-token prompt, tokenization, and counterfactual pass are not reproducible

Location. Manuscript Sections 3.2, 3.3, and 4.4; Algorithm 1.

Problem. Likely issue. The manuscript says that the chat template, leading-whitespace handling, and Yes/No tokenization are “described” in Section 4.4, but that section only says they follow each model’s official inference code. No exact prompts, chat-template expansions, answer token IDs, allowed output variants, invalid-output rule, or per-model parsing rule are given. This omission is critical because $T = 1$ assumes that the first generated token is the answer itself. The text-only counterfactual is also under-specified across seven heterogeneous VLMs: it is unclear whether the image placeholder is removed, retained without embeddings, or replaced; how position IDs are constructed; and whether X_t is exactly the same textual sequence in both streams.

Why it matters. A leading newline or a tokenizer in which “Yes” and “Yes” differ can change both $H(Y_1)$ and the one-token prediction. Removing visual tokens can create an out-of-distribution prompt effect rather than isolate a language prior. These choices can therefore change the gate, the amateur logits, and every reported result.

Suggested fix. Add a per-model implementation table containing the complete rendered prompt, image-placeholder treatment, token IDs/string variants, stopping and parsing rules, and the

construction of the text-only sequence. State whether logits are unrestricted over the vocabulary or restricted/aggregated over Yes/No variants. Release a minimal inference example and verify that the first generated token is an answer token for every evaluated query and model.

4. The significance tests do not test the headline balanced-accuracy differences

Location. Manuscript Section 5.1, especially the exact paired McNemar claims.

Problem. Likely issue. McNemar’s test on per-item correctness tests a difference in ordinary accuracy. The paper’s headline metric is balanced accuracy, which weights the positive and negative classes equally, while the held-out split is imbalanced. Therefore a McNemar p -value does not establish that a reported balanced-accuracy gap is significant. The manuscript also reports several pairwise tests without a stated multiplicity policy or confidence intervals. The quoted 30 versus 10 discordant counts do reproduce a two-sided exact value near $p = 0.0022$, but that calculation remains a test of accuracy.

Why it matters. The text currently attaches significance language directly to balanced-accuracy gaps, so readers may infer support for a different estimand than the one actually tested.

Suggested fix. Use a paired, label-stratified bootstrap or permutation test for balanced accuracy and report confidence intervals for each method difference. State the family of comparisons and apply an appropriate multiplicity correction or clearly label the tests exploratory. McNemar results may remain, but should be described explicitly as tests of ordinary per-item accuracy.

5. Algorithm 1 contradicts the forward-pass accounting

Location. End of manuscript Section 3.2; Algorithm 1; paragraph after the algorithm.

Problem. Definite error. The algorithm computes z_{std} before routing and then recomputes z_{std} at the top of the System 2 loop before selecting the first token. Thus, as written, a routed $T = 1$ query uses the gate pass, a duplicate multimodal pass, and a counterfactual pass: three forward passes, not a reused standard pass plus one counterfactual pass. The System 1 branch also computes another z_{std} after appending the token even when $T = 1$ or the appended token is EOS.

Why it matters. This conflicts with the claimed cost and can cause implementations to disagree. It also weakens statements that the method keeps inference cost “in line,” for which no latency or FLOP measurement is reported.

Suggested fix. Reuse the pre-gate z_{std} for the first decoding decision in both branches. Recompute logits only if another token will be generated. Then report either measured latency or the exact single-token pass count: one multimodal pass for every query plus one text-only pass for the routed fraction (subject to architecture-specific cost).

6. The benchmark scope is described too broadly

Location. Introduction (“touch, align, or support”); Experimental Setup (“spatial alignment and physical contact”); related interpretation throughout.

Problem. Definite error. The cited AMBER specification defines its relation annotation and prompt in terms of *direct contact* between two objects. The 1,664-item relation count is consistent with that source, but alignment and support are not established as separate predicates in the cited benchmark description.

Why it matters. The broader wording overstates the tested relational scope and may lead readers to infer evidence for spatial predicates not evaluated here.

Suggested fix. Describe the evaluated subset consistently as direct-contact relation questions. If a modified AMBER subset actually includes alignment or support, document its construction, counts, labels, and release artifacts explicitly.

7. “Architecture-aware” and “small tuning split” overstate the method

Location. Abstract, contribution list, and manuscript Sections 5.3 and 6.1.

Problem. Unsupported claim. The gate receives no architectural descriptor; it uses a separately supervised threshold for each model, selected by balanced accuracy on 822 of 1,664 examples (49.4% of the benchmark). Calling this split “small” is inaccurate, and calling the method architecture-aware obscures the role of model-specific labeled calibration. The categories “biased” and “structurally sound” are also assigned post hoc from performance patterns rather than from a measured structural property.

Why it matters. The current language makes the method appear to infer architecture-level failure modes automatically, when its operating point is selected separately with substantial labeled data.

Suggested fix. Use “per-model calibrated” or “model-adaptive” instead of “architecture-aware,” replace “small tuning split” with the exact size, and include a calibration-size learning curve. Report a genuinely shared threshold and a calibration-free baseline if cross-model transfer is a central claim.

8. Controls and the LLC baseline need enough detail to support comparisons

Location. Manuscript Sections 4.4, 5.1, and 5.4.

Problem. Likely issue. The shuffled-routing control is reported for only three models, with no number of random repetitions, uncertainty, seed, or label-stratification rule. One random routing can differ substantially in positive/negative composition. The LLC baseline is described only as taking layer 3 and applying the LM head; it is unclear whether the model’s final normalization is applied, where the hidden state is tapped, and why the same absolute layer is comparable across model depths and families.

Why it matters. The random control is used to claim that query selection, rather than intervention frequency, explains “most” of the gain. The near-no-op LLC results may reflect an under-specified or poorly matched intermediate-logit construction rather than a meaningful negative baseline.

Suggested fix. Average many rate-matched random routings, stratify them by ground-truth label and preferably baseline answer, and report confidence intervals. More directly, report CD fix/harm rates by entropy bin and label. For LLC, specify pre/post-normalization, layer indexing, vocabulary head, cache handling, and the rationale or tuning procedure for the amateur layer.

Notation and terminology audit

The main mathematical symbols are introduced consistently: X_v is the visual input, X_t the textual prompt, Y the generated sequence, z_{std} the multimodal logits, z_{cf} the text-only logits, V_{head} the plausibility set, and τ the entropy threshold. I found no initial/final, source/observer, or local/global descriptor conflict of the kind common in coordinate derivations. The consequential drift is conceptual rather than symbolic: $H(Y_1) \geq \tau$ is first defined mathematically as high predictive entropy, but later described as “vision–text conflict,” “Visual Inertia,” and “real trouble.” Those descriptors are not equivalent to the defined statistic and should be replaced or experimentally justified.

Meaningful editorial and presentation issues

- **Results prose, Section 5.2.** “The generation manifold [is] unstable” and “every answer now comes out of a distorted distribution” are not operationally defined or measured. Replace them with the observed recall/specificity shift.
- **Section 5.4.** The statement that the results keep “fluency” in line is unsupported because the evaluation produces one answer token and includes no fluency metric. Remove “fluency” or add an open-ended evaluation.
- **Section 5.1.** “Tracks universal CD+APC within 0.6 percentage points on every model where CD+APC helps” is misleading because EGCD is 3.14 points *above* CD+APC on Qwen2-VL-2B. If the intended claim is one-sided, write “never trails CD+APC by more than 0.57 points where CD+APC improves on greedy.”
- **Submission readiness.** The author block and affiliation remain placeholders. The code and per-item data are promised only upon publication even though they are necessary to check the split, routing, AUC, and statistical claims. Provide an anonymized artifact for review if journal policy permits.

Proofing sweep

The bundled mechanical scan was run on the compiled manuscript, the TeX source, and the bibliography. Its apparent double-period hit was the typeset mathematical ellipsis, and its semicolon-capitalization hits were proper names, acronyms, or list constructions rather than errors. No duplicated-punctuation or malformed-reference hit survived spot-checking. The following high-confidence presentation defects remain:

- **Figures 2–5 and 8.** The plots label the model “Phi-3-Vision-4.2B,” whereas the prose and table use “Phi-3.5-Vision-4.2B.” Regenerate the figures with the exact model name.
- **Figure 6 caption.** It says universal CD+APC falls “up to 10 points” below baseline, but manuscript Table 1 reports a 10.62-point drop for LLaVA-v1.6-Mistral-7B. Use “up to 10.62 points” or “about 11 points.”
- **Abstract and contribution list.** Replace “small tuning split” with “822-example tuning split” unless the calibration set is reduced.

Highest-priority revisions

1. Correct and release the split definition and class counts, then regenerate all metrics from that split.
2. Reframe the mechanistic claims or add direct attention, modality-conflict, and entropy-stratified fix/harm evidence.
3. Specify the exact per-model prompt, answer-token handling, and text-only counterfactual construction.
4. Replace accuracy-based significance claims for balanced-accuracy differences with paired, class-aware uncertainty estimates and tests.

5. Correct Algorithm 1 and report unambiguous compute accounting.

Items requiring external verification

- **Needs external verification.** The AUC values, routed-query percentages, shared-threshold result, shuffled-routing numbers, and unreported discordant-pair counts cannot be reconstructed from the manuscript alone. They require the split IDs, per-item labels, entropies, baseline outputs, and method outputs.
- **Needs external verification.** The claim that image removal isolates a language prior must be checked for each model against the released implementation, because the counterfactual template and image-placeholder behavior are currently unspecified.
- **Needs external verification.** The broad literature statement that relational errors are “generated, not perceived” is stronger than the cited evidence summarized in the manuscript. It should be qualified unless the cited studies rule out encoder/perception errors rather than merely demonstrating decoding- or attention-associated failures.
- **Needs external verification.** Re-check whether wang2025contextfocus supports the surrounding claim about selective interventions in vision-language models: the cited work concerns contextual faithfulness in text-only language models, so its role in that sentence is not evident.